# NOTRE DAME UNIVERSITY

# BANGLADESH

# EEE Lab Report-02

**Course Title: Electrical Circuit Analysis Lab**

**Course Code: EEE-2108**

**Experiment No: 02**

**Name of the Experiment: Finding Q point & designing a circuit for small signal analysis.**

**Date of Experiment: 03-08-23**

**Date of Submission: 10-08-23**

## Submitted by:

**Group: E**

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## Experiment No 2

**Name of the experiment:** Fixing a Q point & designing a circuit for small signal analysis.

**Objective:** Study of the small signal analysis.

**Theory:** Q point or the operating point of a diode, also known as a bias point, or quiescent point is the steady-state DC voltage or current at a specified terminal of an active device. such as a diode or transistor with no input signal applied. In other words, this operating point Q is the intersection where the optimum forward voltage & forward current converge & It's also the point where the diode operates at it's optimum.

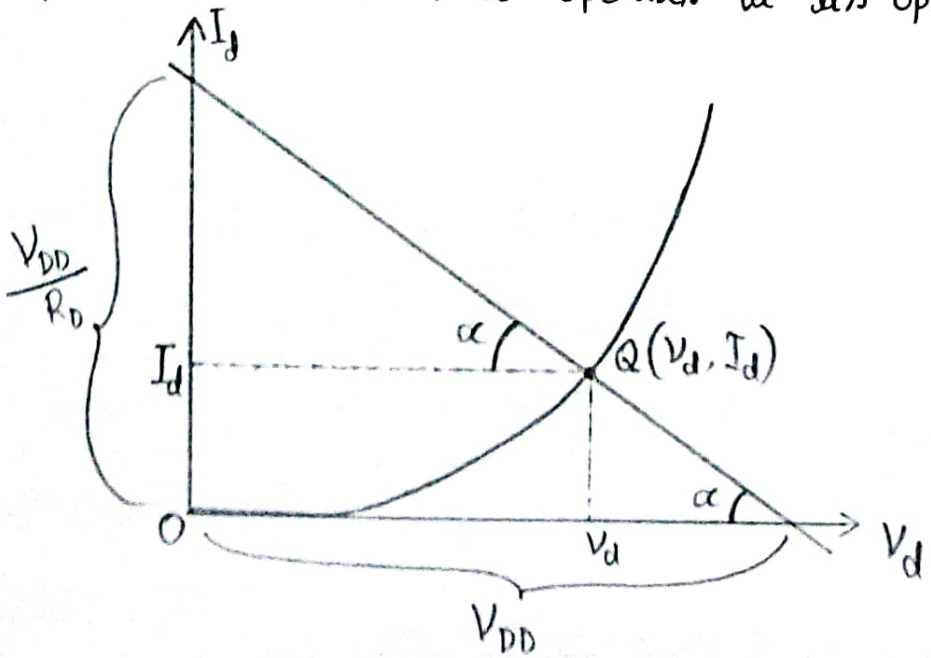



Fig: 2.1

Here,

$$I_d = \frac{V_{DD} - V_d}{R_D}$$

$$\Rightarrow I_d = -\frac{V_d}{R_D} + \frac{V_{DD}}{R_D}$$

$$\Rightarrow I_d = \left(-\frac{1}{R_D}\right) V_d + \frac{V_{DD}}{R_D}$$

Considering this Circuit,

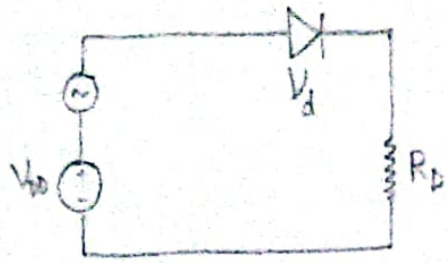



Fig - 2.2

Comparing this equation with $y = mx + c$

$$\tan\alpha = \frac{I_d}{V_{DD} - V_d}$$

$$\Rightarrow \frac{I_d}{V_{DD} - V_d} = \frac{\frac{V_{DD}}{R_D} - I_d}{V_d}$$

$$\Rightarrow V_d I_d = (V_{DD} - V_d)\left(\frac{V_{DD} - I_d R_D}{R_D}\right)$$

$$\Rightarrow V_d I_d R_D = (V_{DD} - V_d)(V_{DD} - I_d R_D)$$

## Equipments:

1. p-n junction diode
2. Resistor
3. Trainer Board.
4. DC Power supply
5. Multimeter.
6. wires
7. Signal generator
8. Digital Storage Oscilloscope.

## Procedures:

1. Connections were made as per the circuit diagram.
2. We measured the value of $V_{DD}$ from the equation putting the value of $I_d$, $R_D$ & $V_d$.
3. Then we measured the maximum & minimum voltage of $V_{DD}$
4. After that we used the average of $V_{DD}$ in the circuit & got the maximum & minimum voltage of $V_D$
5. We simply measured the maximum & minimum current across diode by using the value of $V_{DD}$, $V_d$ & $R_D$
6. Then we calculated M by using this formula,

$$M = \frac{I_{d_{max}} - I_{d_{min}}}{V_{d_{max}} - V_{d_{min}}}$$

7. Signals were shown in the oscilloscope.

Group-E

# Experiment-2

Fixing a Q point and designing a circuit for small signal analysis

Given, Q(0.635V, 2.965mA)

$V_d = 0.635V$ , $R_D = 1k\Omega$

$I_d = 2.965mA$

$V_D = ?$

$$V_d I_d R_D = (V_{DD} - V_d)(V_{DD} - I_d R_D)$$

$$\Rightarrow 0.635 \times 2.965 \times 1 = (V_{DD} - 0.635)(V_{DD} - 2.965)$$

$$\Rightarrow 1.883 = V_{DD}^2 - 0.635 V_{DD} - 2.965 V_{DD} + 1.883$$

$$\Rightarrow V_{DD}^2 = 3.6 V_{DD}$$

$$\Rightarrow V_{DD} = 3.6V$$

$V_{dc} + V_{ac}$

$V_{dd}$

$V_{dmax} = 0.66V$ , $V_{DDmax} = 3.6 + 0.18 = 3.78 \approx 3.8V$

$V_{dmin} = 0.62V$ , $V_{DDmin} = 3.6 - 0.18 = 3.42 \approx 3.4V$

$$I_{dmax} = \frac{V_{DDmax} - V_{dmax}}{R_D} = \frac{3.8 - 0.66}{1} = 3.14mA$$

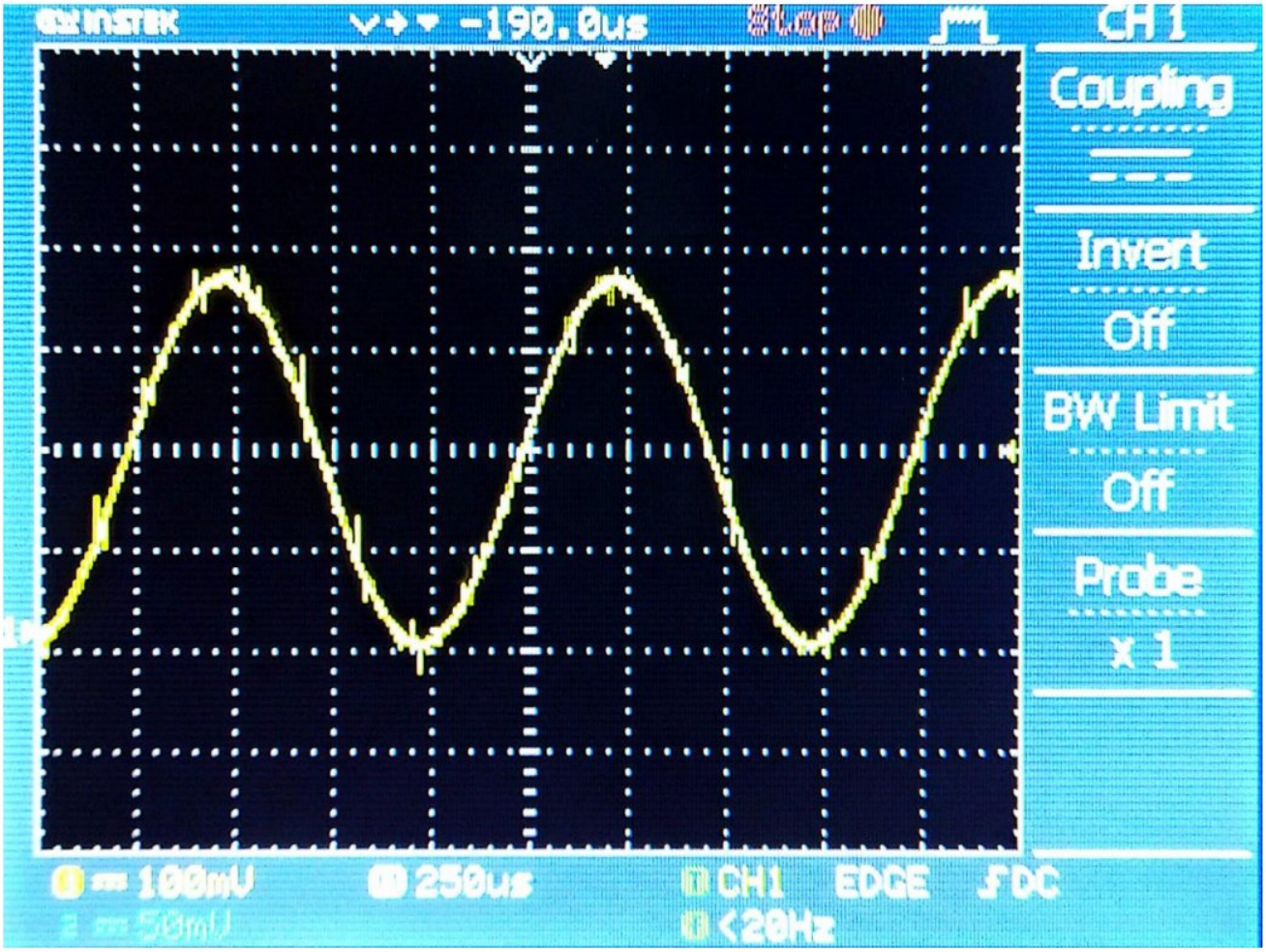
$$I_{dmin} = \frac{V_{DDmin} - V_{dmin}}{R_D} = \frac{3.4 - 0.62}{1} = 2.78mA$$

$$m = \frac{I_{dmax} - I_{dmin}}{V_{dmax} - V_{dmin}} = \frac{3.14 - 2.78}{0.66 - 0.62}$$

$$= 9mA/V$$

03/08/23

# Oscilloscope Input Graph:




Graph: Input Signal Graph through oscilloscope

## Discussion:

1. When supplying the AC ~~ex~~ Signal. It's required a steady DC Current or voltage or to operate correctly.

2. We have to find Q point so that we can consider a Linear part in the $I_d$ Vs $V_d$ ~~gp~~ graph

3. Avoid overloading & short circuit

4. Follow guidelines & Manual.